Autonomous Support Resolution Agent

Agentic AI System with Multi-Step Tool Execution

Presented by: Vivek Kumar

The Problem

High Volume

Thousands of support tickets daily

Repetitive Queries

80% are common questions

High Cost

Manual processing is expensive

Slow Response

Hours to days for resolution

The Solution

Autonomous AI Agent

Not just a chatbot — a system that can:

- Think and reason about problems
- Make intelligent decisions
- Execute actions using real tools

Key Features

Multi-Step Reasoning

ReAct loop for complex logic

Tool Execution

Real actions, not just responses

Confidence Scoring

Self-awareness of certainty

Failure Handling

Retry and escalation logic

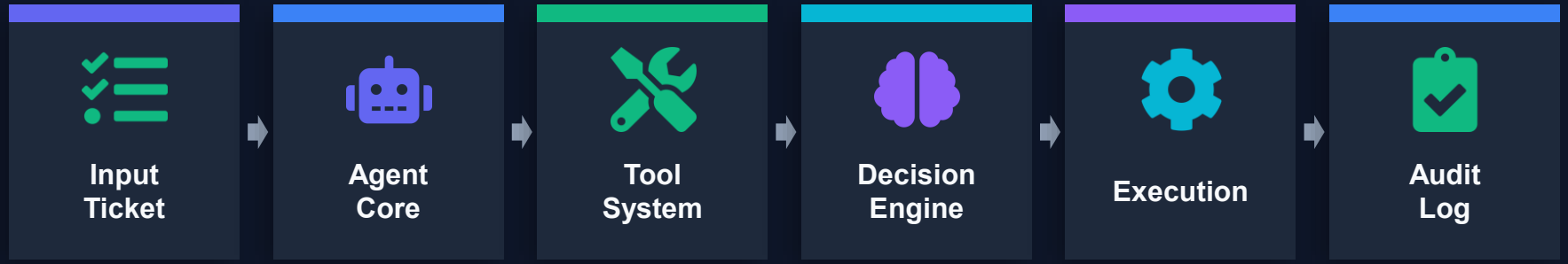
Audit Logging

Full transparency & traceability

Parallel Processing

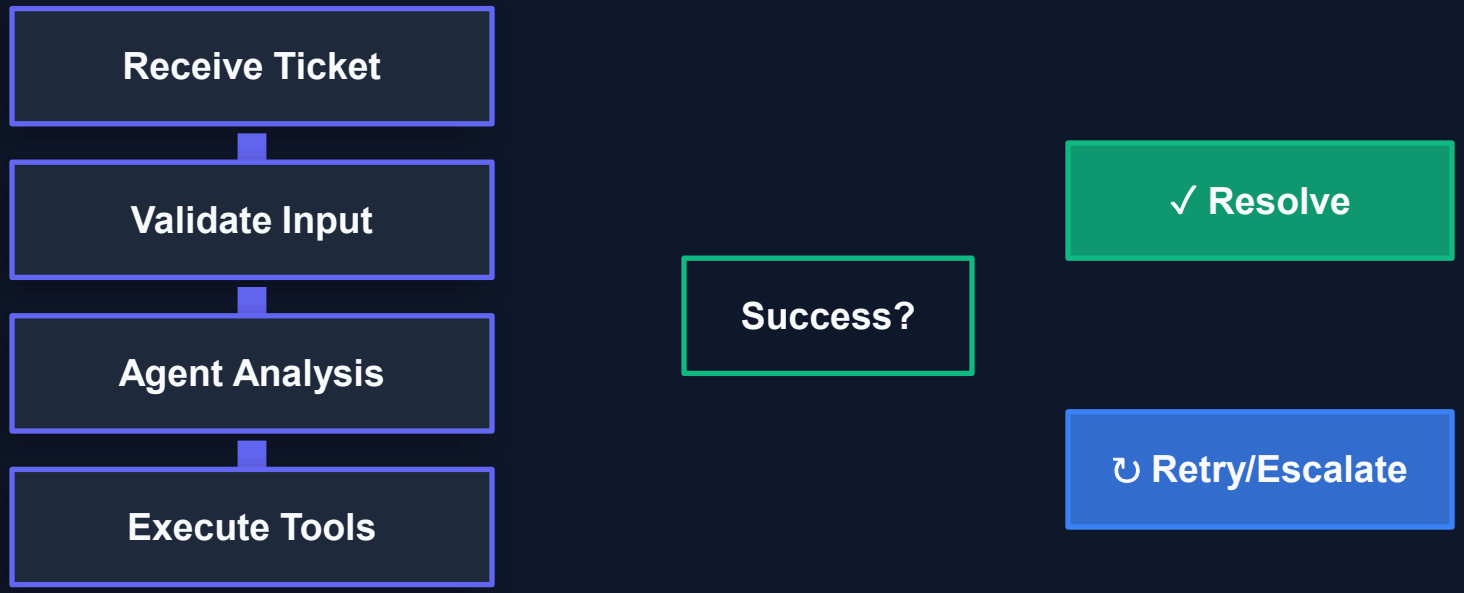
Handle multiple tickets at once

System Architecture



Continuous feedback loop for learning and improvement

Workflow


Receive Ticket
Validate Input
Agent Analysis
Execute Tools
Success?
✓ Resolve
↻ Retry/Escalate

How It Works

1. Receive and parse support ticket
2. Validate ticket format and requirements
3. Start agent reasoning loop (ReAct)
4. Call appropriate tools for data/actions
5. Evaluate results and decide next step
6. Execute final action (resolve/escalate)
7. Log all steps for audit trail

Tool System

Read Tools

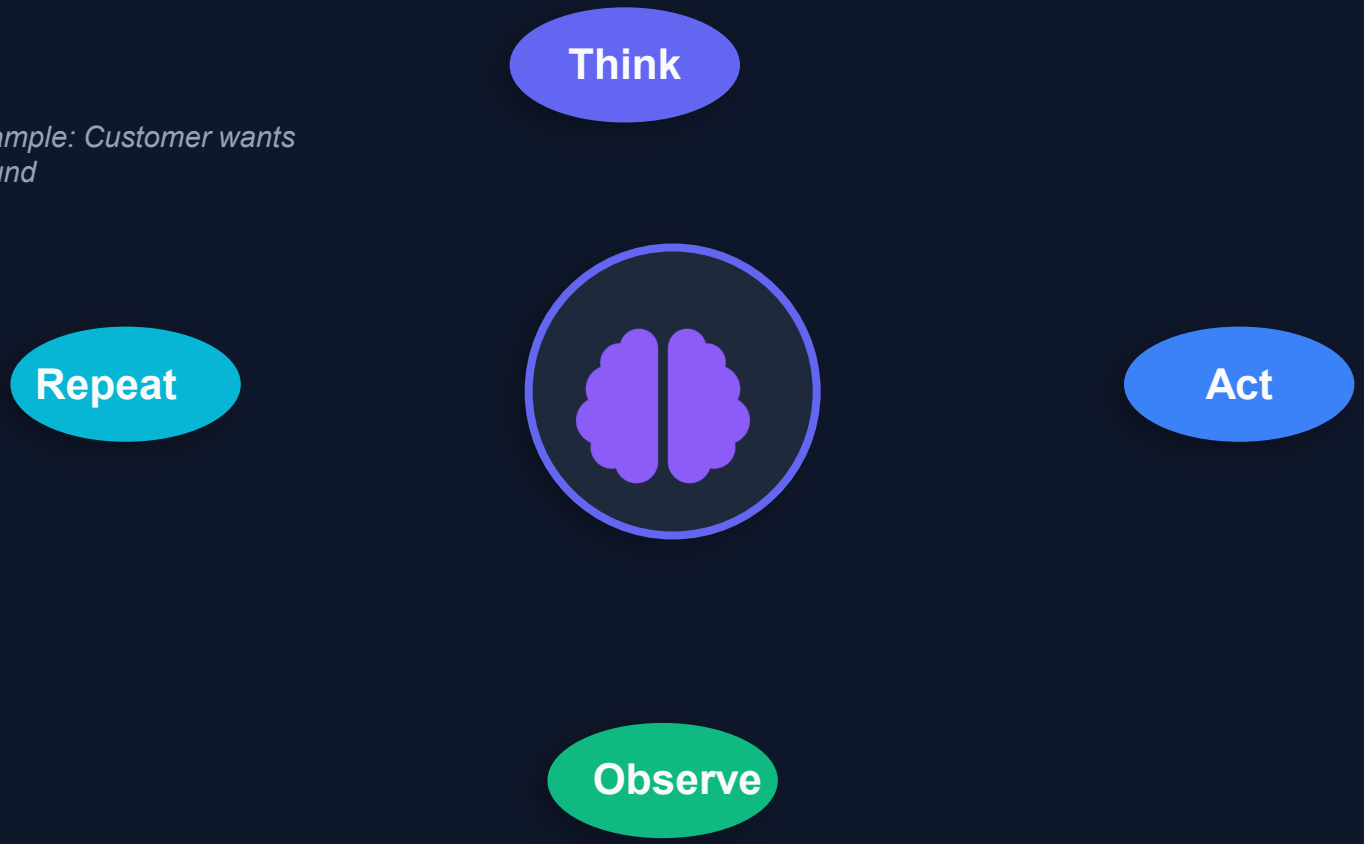
- get_customer
- get_order
- search_knowledge_base

Action Tools

- check_refund_eligibility
- issue_refund
- send_reply
- escalate

Agent Reasoning: ReAct Loop

Example: Customer wants refund



Demo Execution

```
$ python agent.py --ticket ticket_001.json

> Starting agent...
> Ticket ID: #12345
> Issue: Refund request

[THINK] Need customer and order data
[ACT] Calling get_customer(12345)
[OBSERVE] Customer verified: VIP status
[ACT] Calling check_refund_eligibility(order_789)
[OBSERVE] Eligible: Yes (within 30 days)
[ACT] Calling issue_refund(order_789, amount: $49.99)
[OBSERVE] Refund processed successfully
[ACT] Calling send_reply("Refund processed")

✓ Ticket resolved in 2.3 seconds
```

Code Implementation

```
class SupportAgent:
    def __init__(self, tools):
        self.tools = tools
        self.max_iterations = 10

    def solve(self, ticket):
        for i in range(self.max_iterations):
            thought = self.think(ticket, history)
            action = self.decide_action(thought)

            if action.type == "resolve":
                return self.resolve(action)

            result = self.execute_tool(action)
            self.log(thought, action, result)
            history.append((action, result))

        return self.escalate(ticket)
```

Failure Handling

Retry Mechanism
Automatic retry with exponential backoff

Input Validation
Verify data before tool execution

Graceful Degradation
Fallback to simpler strategies

Smart Escalation
Route complex cases to humans

Audit Logging

```
{
  "ticket_id": "12345",
  "timestamp": "2024-01-15T10:30:00Z",
  "agent_version": "v2.1",
  "steps": [
    {"action": "get_customer", "result": "success"},
    {"action": "check_refund", "result": "eligible"},
    {"action": "issue_refund", "result": "completed"}
  ],
  "resolution": "resolved",
  "duration_ms": 2300,
  "confidence": 0.95
}
```

Parallel Processing

Handle Multiple Tickets Simultaneously

- Process 10-20 tickets in parallel
- Shared tool pool for efficiency

Sequential: 100 tickets/hour → Parallel: 1000+ tickets/hour

10x faster than sequential

Results

20	12	5	3
Total Tickets	Resolved	Escalated	Failed

60% Full Automation Rate • 25% Smart Escalation

Chatbot vs AI Agent

Traditional Chatbot

- ❌ Only responds to queries
- ❌ No real actions
- ❌ Rule-based logic
- ❌ Can't learn
- ❌ Limited context
- ❌ No decision making

AI Agent

- ✓ Autonomous decision making
- ✓ Executes real actions
- ✓ Multi-step reasoning
- ✓ Learns from outcomes
- ✓ Full context awareness
- ✓ Tool-based execution

Future Enhancements

API Integration

Real-time CRM/ticketing system sync

Database Layer

Persistent storage & analytics

Web Dashboard

Real-time monitoring & control

Multi-Agent System

Specialized agents for different domains

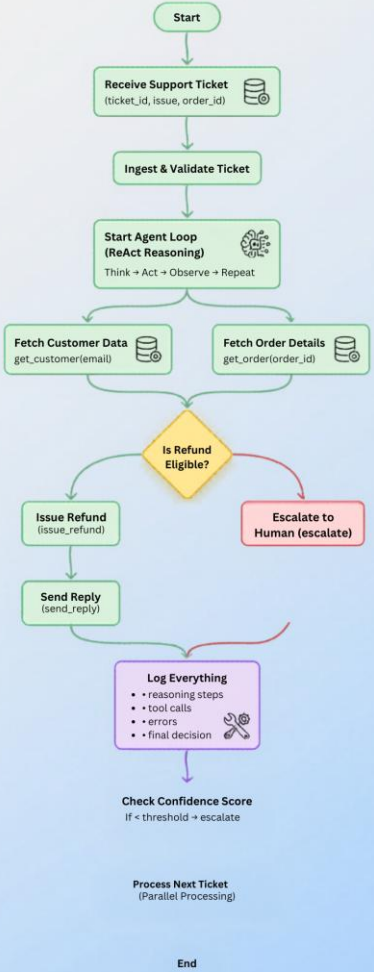
Key Takeaways

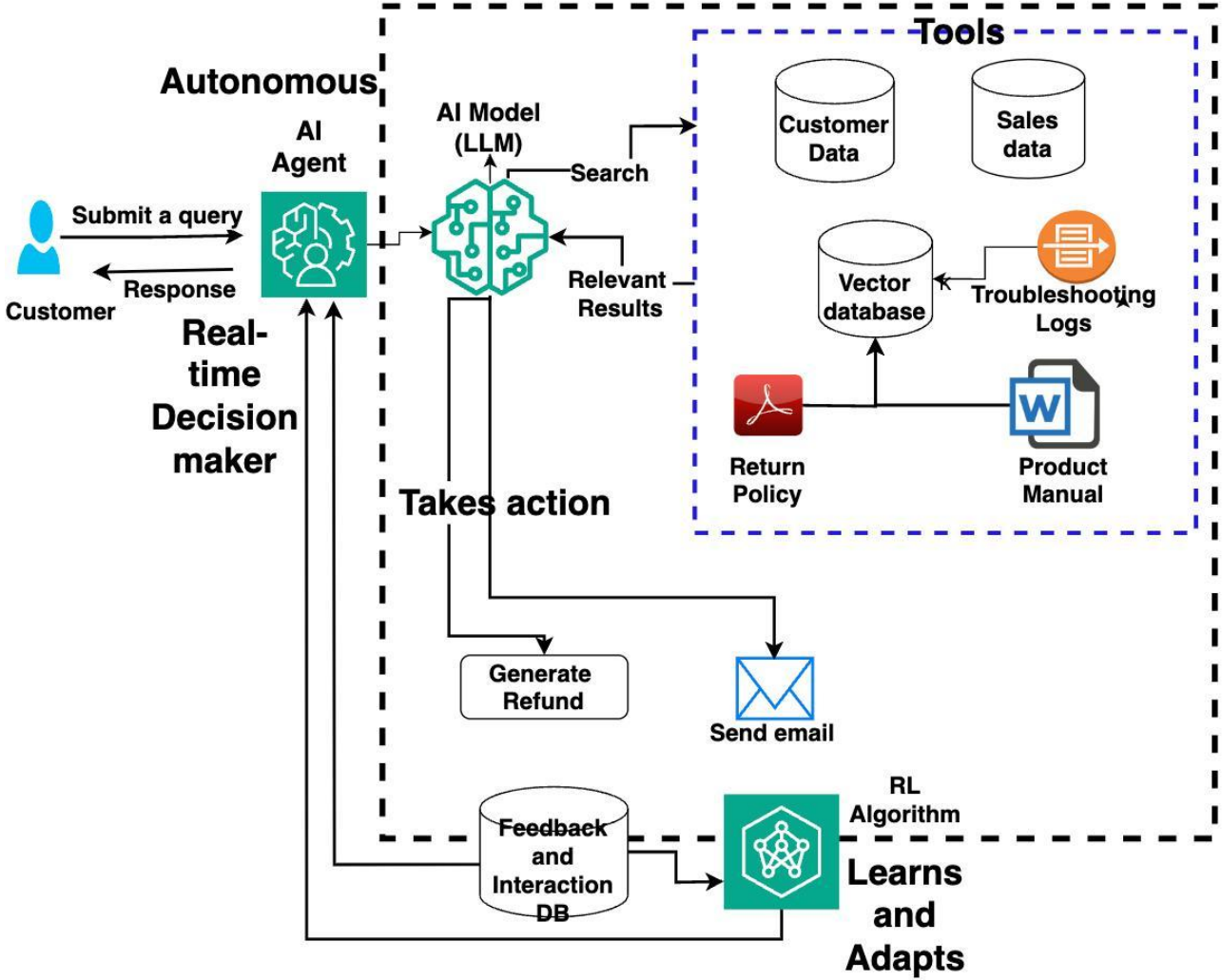
Autonomous AI System

- ✓ Goes beyond chatbots with real decision-making
- ✓ Multi-step reasoning with tool execution
- ✓ Production-ready with failure handling
- ✓ Scalable and transparent

Ready to transform support automation

Autonomous Support Resolution Agent – Flowchart




Autonomous
AI Agent
Submit a query
Response
Customer
Real-time Decision maker
AI Model (LLM)
Search
Relevant Results
Tools
Customer Data
Sales data
Vector database
Troubleshooting Logs
Return Policy
Product Manual
Takes action
Generate Refund
Send email
Feedback and Interaction DB
RL Algorithm
Learns and Adapts

SUBJECT *

Order delayed, want refund

DESCRIPTION *

My order was supposed to be delivered 5 days ago but it is still not delivered. I would like to request a refund for this order.

EMAIL *

vk1124629@gmail.com

ORDER ID

84886646788

PRIORITY

High

CATEGORY

Billing

Cancel

Submit & Run Agent

Tickets

4 total tickets

+ New Ticket

Search tickets...

All Statuses

All Priorities

TKT-20260419-14FE07 High • IN PROGRESS

Order delayed, want refund

My order was supposed to be delivered 5 days ago but it is still not delivered. I would like to request a refund for this...

vk1124629@gmail.com Apr 19, 01:17 PM

TKT-20260419-456A13 High • RESOLVED

Order delayed, want refund

My order was supposed to be delivered 5 days ago but it is still not delivered. I would like to request a refund for this...

vk1124629@gmail.com Apr 19, 01:10 PM

AI Confidence 50%

TKT-20260419-E1FD4E High • RESOLVED

pymrnt

my pyment done but reflected

vk1124629@gmail.com Apr 19, 12:43 PM

AI Confidence 50%

TKT-20260419-8AA93C Medium • RESOLVED

order fail

payment fails i

vk1124629@gmail.com Apr 19, 12:41 PM

AI Confidence 50%

Tickets
Manage & create

Audit Logs
Real-time activity

audit_log

3 entries

6:11:22 PM	resolved	TKT-20260419-8AA93C	order fail	50%
6:13:18 PM	resolved	TKT-20260419-E1FD4E	pymrnt	50%
6:40:47 PM	resolved	TKT-20260419-456A13	Order delayed, want refund	50%

Latest Entry

JSON


```
{
  "ticket_id": "TKT-20260419-456A13",
  "subject": "Order delayed, want refund",
  "user_email": "vk1124629@gmail.com",
  "status": "resolved",
  "final_decision": "unknown",
  "confidence_score": 0.5,
  "reasoning": [
    "Customer is requesting a refund. I need to verify their identity, check the order, and process the refund if eligible.",
```

Dashboard

AI-Powered

Autonomous Support Resolution Agent — real-time overview

Refresh

TOTAL TICKETS	RESOLVED	ESCALATED	FAILED
3	3	0	0
• 0 in progress	• 100% rate	• Needs human review	• Agent errors

Distribution

Ticket breakdown

Resolved	3
Escalated	0
Failed	0
Pending	0

Resolution Rate

Overall performance

100%

of tickets resolved autonomously

0% 50% 100%

Activity

Latest updates

View All →

Order delayed, want refund TKT-20260419-456A13	01:10 PM	RESOLVED
pymrnt TKT-20260419-E1FD4E	12:43 PM	RESOLVED
order fail TKT-20260419-8AA93C	12:41 PM	RESOLVED

Recent Tickets

3 shown

See all tickets →

TKT-20260419-456A13 High RESOLVED

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AI Confidence 50%

SupportAgent
AI-Powered

NAVIGATION

Dashboard
Overview & stats

Tickets
Manage & create

Audit Logs
Real-time activity

TKT-20260419-456A13 • RESOLVED

Order delayed, want refund

Ticket Information

Description	My order was supposed to be delivered 5 days ago but it is still not delivered. I would like to request a refund for this order.
Customer	vk1124629@gmail.com
Priority	high
Category	billing
Team	—
Created	4/19/2026, 1:10:46 PM
Resolved	4/19/2026, 6:40:47 PM

FINAL ACTION

Unknown

AI CONFIDENCE

50%

AI Reasoning Chain

10

master / Autonomous-Support-Resolution-Agent- / frontend /

vivekdeveloper20 Complete UI redesign: premium glassmorphic dashboard with animation... ce8406e · 1 hour ago History

Name	Last commit message	Last commit date
..		
public	Initial commit from local workspace	4 hours ago
src	Complete UI redesign: premium glassmorphic dashboard with animation...	1 hour ago
.gitignore	Initial commit from local workspace	4 hours ago
index.html	Complete UI redesign: premium glassmorphic dashboard with animation...	1 hour ago
package-lock.json	Initial commit from local workspace	4 hours ago
package.json	Initial commit from local workspace	4 hours ago
tsconfig.json	Initial commit from local workspace	4 hours ago
vite.config.js	Initial commit from local workspace	4 hours ago

master / Autonomous-Support-Resolution-Agent- / Ai-Ticket-Agent-master / ai_ticket_agent /

vivekdeveloper20 Initial commit from local workspace 346e3c · 4 hours ago History

Name	Last commit message	Last commit date
..		
sub_agents	Initial commit from local workspace	4 hours ago
tools	Initial commit from local workspace	4 hours ago
__init__.py	Initial commit from local workspace	4 hours ago
agent.py	Initial commit from local workspace	4 hours ago
database.py	Initial commit from local workspace	4 hours ago
models.py	Initial commit from local workspace	4 hours ago
prompt.py	Initial commit from local workspace	4 hours ago

vivekdeveloper20 / Autonomous-Support-Resolution-Agent-

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

master / Autonomous-Support-Resolution-Agent- / backend / data /

vivekdeveloper20 feat: add demo data, update tools, add root .gitignore 0cc8301 · 1 hour ago History

Name	Last commit message	Last commit date
..		
customers.json	feat: add demo data, update tools, add root .gitignore	1 hour ago
knowledge-base.md	feat: add demo data, update tools, add root .gitignore	1 hour ago
orders.json	feat: add demo data, update tools, add root .gitignore	1 hour ago
products.json	feat: add demo data, update tools, add root .gitignore	1 hour ago
tickets.json	feat: add demo data, update tools, add root .gitignore	1 hour ago

master / Autonomous-Support-Resolution-Agent- / Ai-Ticket-Agent-master /

vivekdeveloper20 Add additional flowchart images ae7cc82 · 3 hours ago History

Name	Last commit message	Last commit date
..		
ai_ticket_agent	Initial commit from local workspace	4 hours ago
images	Add additional flowchart images	3 hours ago
tests	Initial commit from local workspace	4 hours ago
.gitignore	Initial commit from local workspace	4 hours ago
DASHBOARD_GUIDE.md	Initial commit from local workspace	4 hours ago
README.md	Add visual flowchart images to README documentation	3 hours ago
SLACK_SETUP.md	Initial commit from local workspace	4 hours ago
dashboard.py	Initial commit from local workspace	4 hours ago
.env.example	Initial commit from local workspace	4 hours ago
init_database.py	Initial commit from local workspace	4 hours ago
poetry.lock	Initial commit from local workspace	4 hours ago
pyproject.toml	Initial commit from local workspace	4 hours ago
run.py	Initial commit from local workspace	4 hours ago

README.md

master / Autonomous-Support-Resolution-Agent- / Ai-Ticket-Agent-master / ai_ticket_agent / tools /

vivekdeveloper20 Initial commit from local workspace 346e3c · 4 hours ago History

Name	Last commit message	Last commit date
..		
__init__.py	Initial commit from local workspace	4 hours ago
email_collector.py	Initial commit from local workspace	4 hours ago
email_sender.py	Initial commit from local workspace	4 hours ago
knowledge_base.py	Initial commit from local workspace	4 hours ago
notification_sender.py	Initial commit from local workspace	4 hours ago
problem_analyzer.py	Initial commit from local workspace	4 hours ago
resolution_tracker.py	Initial commit from local workspace	4 hours ago
slack_handlers.py	Initial commit from local workspace	4 hours ago
team_router.py	Initial commit from local workspace	4 hours ago
ticket_manager.py	Initial commit from local workspace	4 hours ago

master / Autonomous-Support-Resolution-Agent- / backend /

vivekdeveloper20 feat: add demo data, update tools, add root .gitignore 0cc8301 · 1 hour ago History

Name	Last commit message	Last commit date
..		
__pycache__	second	1 hour ago
data	feat: add demo data, update tools, add root .gitignore	1 hour ago
logs	second	1 hour ago
.env	second	1 hour ago
.env.example	Initial commit from local workspace	4 hours ago
audit_logger.py	Initial commit from local workspace	4 hours ago
database.py	Initial commit from local workspace	4 hours ago
main.py	Initial commit from local workspace	4 hours ago
models.py	Initial commit from local workspace	4 hours ago
react_agent.py	Initial commit from local workspace	4 hours ago
requirements.txt	Complete UI redesign: premium glassmorphic dashboard with animation...	1 hour ago
seed_demo.py	Initial commit from local workspace	4 hours ago
support_agent.db	Initial commit from local workspace	4 hours ago
tools.py	feat: add demo data, update tools, add root .gitignore	1 hour ago

Thank You

Name: Vivek Kumar

Contact No: +91 8434487300

Email: **vk1124629@gmail.com**

LinkedIn: https://www.linkedin.com/in/vivekkumar56/

Portfolio: https://vivekg.vercel.app/

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